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Information: Updating design information... (UID-85)
Warning: Design 'riscv_core_timing' contains 1 high-fanout nets. A fanout number of 1000 will be used for delay calculations↙
involving these nets. (TIM-134)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : riscv_core_timing
Version: U-2022.12-SP2
Date   : Mon Apr 21 23:49:00 2025
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 # A fanout number of 1000 was used for high fanout net computations.

Operating Conditions: typical   Library: typical
Wire Load Model Mode: top

  Startpoint: RISCV_Core/r4/instr_F2DE_reg[9]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: RISCV_Core/fetchStage/pc_reg[27]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Point                                                                                                   Incr       Path
↙
-----------------------------------------------------------------------------------------------------------------------------↙
-
  clock clk (rise edge)                                                                                   0.00       0.00
  clock network delay (ideal)                                                                             0.00       0.00
  RISCV_Core/r4/instr_F2DE_reg[9]/CK (DFFHQX4)                                                            0.00 #     0.00↙
r
  RISCV_Core/r4/instr_F2DE_reg[9]/Q (DFFHQX4)                                                             0.13       0.13↙
f
  RISCV_Core/r4/instr_F2DE[9] (registerEM1)                                                               0.00       0.13↙
f
  RISCV_Core/U334/Y (NOR3X1)                                                                              0.14       0.27↙
r
  RISCV_Core/U276/Y (AOI31X1)                                                                             0.07       0.34↙
f
  RISCV_Core/U183/Y (CLKINVX1)                                                                            0.09       0.43↙
r
  RISCV_Core/U180/Y (CLKINVX3)                                                                            0.05       0.47↙
f
  RISCV_Core/U30/Y (NAND2BX1)                                                                             0.09       0.56↙
r
  RISCV_Core/U179/Y (NOR2X2)                                                                              0.05       0.61↙
f
  RISCV_Core/U176/Y (AOI21X4)                                                                             0.09       0.70↙
r
  RISCV_Core/U175/Y (AND2X4)                                                                              0.08       0.79↙
r
  RISCV_Core/U53/Y (NAND3X2)                                                                              0.05       0.83↙
f
  RISCV_Core/U255/Y (AOI2BB1X4)                                                                           0.16       0.99↙
f
  RISCV_Core/U256/Y (NAND4BX2)                                                                            0.11       1.09↙
r
  RISCV_Core/U8/Y (NOR2X4)                                                                                0.05       1.14↙
f
  RISCV_Core/U244/Y (NAND2X2)                                                                             0.07       1.21↙
r
  RISCV_Core/U72/Y (INVX2)                                                                                0.03       1.24↙
f
  RISCV_Core/U145/Y (NOR2X2)                                                                              0.06       1.30↙
r
  RISCV_Core/U296/Y (AND3X4)                                                                              0.16       1.45↙
r
  RISCV_Core/executeStage/forwardB[3] (execute)                                                           0.00       1.45↙
r
  RISCV_Core/executeStage/U745/Y (NOR2X1)                                                                 0.05       1.50↙
f
  RISCV_Core/executeStage/U953/Y (BUFX4)                                                                  0.10       1.60↙
f
  RISCV_Core/executeStage/U407/Y (NOR2BX4)                                                                0.11       1.72↙
f
  RISCV_Core/executeStage/U1577/Y (AOI222X1)                                                              0.22       1.93↙
r
  RISCV_Core/executeStage/U1278/Y (NAND4X1)                                                               0.08       2.01↙
f
  RISCV_Core/executeStage/U1791/Y (OAI2BB2X4)                                                             0.13       2.14↙
f
  RISCV_Core/executeStage/ALU/alu_src2[9] (riscv_alu)                                                     0.00       2.14↙
f
  RISCV_Core/executeStage/ALU/U418/Y (XOR2X4)                                                             0.13       2.27↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/B[9] (adder_WIDTH32_1)                                            0.00       2.27↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/B[9] (adder_WIDTH32_1_DW01_add_0_DW01_add_2)  0.00       2.27↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U229/Y (NOR2X4)                               0.08       2.36↙
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r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U209/Y (OAI21X4)                                  0.04       2.39↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U31/Y (AOI21X2)                                   0.11       2.51↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U186/Y (OAI21X2)                                  0.06       2.56↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U16/Y (INVX4)                                     0.04       2.60↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U150/Y (OAI21X2)                                  0.05       2.65↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U115/Y (AOI22X2)                                  0.13       2.78↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U143/Y (NAND3X2)                                  0.05       2.83↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U129/Y (OAI2BB1X4)                                0.11       2.94↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U130/Y (NAND2X2)                                  0.05       2.99↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U122/Y (OAI211X2)                                 0.05       3.04↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U119/Y (NAND2X2)                                  0.07       3.12↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U258/Y (INVX4)                                    0.02       3.14↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U183/Y (NAND2X2)                                  0.04       3.18↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U176/Y (OAI2BB1X4)                                0.10       3.28↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U96/Y (NAND2X2)                                   0.03       3.31↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U178/Y (INVX2)                                    0.04       3.35↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U170/Y (NAND2BX2)                                 0.04       3.39↙
f
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/U235/Y (NAND2X2)                                  0.06       3.44↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/add_1_root_add_75_2/SUM[29] (adder_WIDTH32_1_DW01_add_0_DW01_add_2)   0.00       3.44↙
r
  RISCV_Core/executeStage/ALU/ALU_Adder/sum[29] (adder_WIDTH32_1)                                             0.00       3.44↙
r
  RISCV_Core/executeStage/ALU/U435/Y (AOI222X2)                                                               0.06       3.50↙
f
  RISCV_Core/executeStage/ALU/U242/Y (NAND2X4)                                                                0.10       3.60↙
r
  RISCV_Core/executeStage/ALU/alu_out[29] (riscv_alu)                                                         0.00       3.60↙
r
  RISCV_Core/executeStage/U582/Y (CLKINVX3)                                                                   0.04       3.64↙
f
  RISCV_Core/executeStage/U374/Y (NAND4BX2)                                                                   0.09       3.73↙
r
  RISCV_Core/executeStage/U1284/Y (NOR2X4)                                                                    0.04       3.77↙
f
  RISCV_Core/executeStage/U1283/Y (NAND2X4)                                                                   0.06       3.83↙
r
  RISCV_Core/executeStage/U606/Y (INVX4)                                                                      0.02       3.85↙
f
  RISCV_Core/executeStage/U811/Y (NOR2X2)                                                                     0.08       3.93↙
r
  RISCV_Core/executeStage/U687/Y (NAND2X2)                                                                    0.04       3.97↙
f
  RISCV_Core/executeStage/U672/Y (NAND2X2)                                                                    0.07       4.04↙
r
  RISCV_Core/executeStage/U700/Y (OAI2BB1X4)                                                                  0.05       4.09↙
f
  RISCV_Core/executeStage/needToFlush (execute)                                                               0.00       4.09↙
f
  RISCV_Core/decodeStage/needToFlush (decode)                                                                 0.00       4.09↙
f
  RISCV_Core/decodeStage/U5/Y (NAND2X4)                                                                       0.06       4.14↙
r
  RISCV_Core/decodeStage/U6/Y (NAND2X4)                                                                       0.04       4.18↙
f
  RISCV_Core/decodeStage/U25/Y (INVX4)                                                                        0.05       4.22↙
r
  RISCV_Core/decodeStage/U48/Y (OR2X4)                                                                        0.09       4.32↙
r
  RISCV_Core/decodeStage/rs1[1] (decode)                                                                      0.00       4.32↙
r
  RISCV_Core/U20/Y (CLKINVX8)                                                                                 0.04       4.36↙
f
  RISCV_Core/U237/Y (OAI2BB2X4)                                                                               0.10       4.47↙
f
  RISCV_Core/U33/Y (AND3X4)                                                                                   0.11       4.58↙
f
  RISCV_Core/U32/Y (NAND3X2)                                                                                  0.06       4.63↙
r
  RISCV_Core/U113/Y (AOI21X2)                                                                                 0.05       4.69↙
f
  RISCV_Core/U633/Y (AOI222X2)                                                                                0.23       4.91↙
r
  RISCV_Core/U142/Y (NAND2X2)                                                                                 0.04       4.95↙
f
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  RISCV_Core/U9/Y (CLKINVX3)                                  0.04       4.99↙
r
  RISCV_Core/U338/Y (NAND4X2)                                 0.05       5.05↙
f
  RISCV_Core/U339/Y (INVX4)                                   0.07       5.11↙
r
  RISCV_Core/U776/Y (INVX4)                                   0.04       5.15↙
f
  RISCV_Core/fetchStage/instr_stall (IF)                      0.00       5.15↙
f
  RISCV_Core/fetchStage/U29/Y (OAI21X4)                       0.11       5.26↙
r
  RISCV_Core/fetchStage/U31/Y (INVX4)                         0.05       5.31↙
f
  RISCV_Core/fetchStage/U148/Y (CLKINVX8)                     0.05       5.36↙
r
  RISCV_Core/fetchStage/U87/Y (INVX2)                         0.03       5.39↙
f
  RISCV_Core/fetchStage/U26/Y (OAI2BB2XL)                     0.11       5.50↙
f
  RISCV_Core/fetchStage/pc_reg[27]/D (DFFRXL)                 0.00       5.50↙
f
  data arrival time                                                      5.50

  clock clk (rise edge)                                       5.60       5.60
  clock network delay (ideal)                                 0.00       5.60
  RISCV_Core/fetchStage/pc_reg[27]/CK (DFFRXL)                0.00       5.60↙
r
  library setup time                                         -0.10       5.50
  data required time                                                     5.50
↙
--------------------------------------------------------------------------------↙
-
  data required time                                                     5.50
  data arrival time                                                     -5.50
↙
--------------------------------------------------------------------------------↙
-
  slack (MET)                                                            0.00


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